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P. M. Cann

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P.M. CANN
Imperial College of Science, Technology and Medicine
Tribology Section
London SW7 2BX, England, UK

Many grease lubricated roller bearings operate in the starved elastohydrodynamic (EHL) regime where there is a limited supply of lubricant to the contact (1). Under these conditions the film thickness drops to a fraction of the fully flooded value (2) and, thus, it is difficult to predict lubrication performance, or bearing life, from conventional EHL models. In this regime film thickness depends on the ability of the grease to replenish the track rather than the usual EHL considerations. The conventional view of grease lubrication is that base oil bleeds from the bulk reservoir close to the track, replenishing the inlet and forming a fluid EHL film (3). Resupply, under starved conditions, will thus depend on both operating conditions and grease parameters. The aim of this paper is to evaluate the influence of these parameters on starved lubrication in a rolling contact. Starved film thickness has been measured for a series of greases and the results have been compared to the fully flooded values. These show that the degree of starvation increases with increasing rolling speed, base oil viscosity and thickener content but decreases at higher temperatures. In many cases an increase in absolute film thickness is obtained when moving from high viscosity base oil to a low one, this result is the reverse of normally accepted EHL rules. Taking the fully flooded film thickness as a guide to lubrication performance is therefore not valid as grease film thickness in the starved regime is determined by local replenishment rather than bulk rheological properties.

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EHL; Grease; Rolling Element Bearings

GREASE LUBRICATION OF ROLLING CONTACTS

The primary role of a grease in rolling element bearings is to form a lubricating film sufficient to separate the rubbing surfaces thus minimizing friction and prolonging bearing life. The thickness and properties of the film are thus of fundamental importance in determining the operation and life of a bearing.

Rolling element bearings operate in the elastohydrodynamic (EHL) regime of high loads and relatively high entrainment speeds. Much is known about this regime for conventional oils and it is a relatively simple procedure to compute the film thickness if the contact conditions and lubricant physical properties are known *(5)*. Grease film thickness can also be predicted from conventional rheological properties if a continued flow of bulk lubricant is assumed *(6)*, *(7)*. This condition can occur in some applications, however it is generally undesirable as high torque, excessive churning of the grease, high operating temperatures and failure can result.

If the fully flooded condition is not maintained then the contact starves and the film thickness is reduced *(2)*, *(4)*. Grease has a yield stress and once it has been pushed aside by the rolling action of the ball it will not readily flow back to replenish the contact but forms ridges of worked grease at either side of the track *(4)*. The accepted view of grease behavior is that it acts as a reservoir releasing oil into the rolled track and thus replenishing the contact *(3)*. This model implies that the lubricant film is composed of base oil alone and that the thickness is determined by the base oil properties and the efficiency of supply. It is thus analogous to starved fluid film lubrication.

In this review, grease lubrication has been divided into two distinct supply conditions: continued bulk flow to the inlet and restricted local (base oil) flow from the reservoir at the side of the track. The fully flooded condition is generally undesirable as it leads to high energy losses. The limited supply condition ensures low friction but can lead to very thin films and it is difficult to predict performance from current lubrication models. It is likely however that in an operating bearing the actual mechanism will vary with the running conditions and bearing/grease properties. In a bearing there is no mechanism to continually force grease into the rolling track, however there are some aspects of bearing design and operation (spin, vibration, cage effects) that could promote intermittent flow of bulk grease into the track *(8)*. However for the most severe lubrication conditions it is the supply and film formation mechanisms which occur in the absence of external redistribution of the grease which are important. Earlier work *(2), (4)* has shown that in experiments where the grease supply is not maintained, the film thickness is both time and test condition dependent. With continued rolling, at constant speed, the film thickness drops as the inlet is progressively starved. After extended running times, a steady-state film thickness is usually reached *(9)* which implies that an equilibrium flow condition has been achieved.

There is little direct evidence concerning the film loss and replenishment mechanisms in the starved grease regime, however fluid film models *(10), (11)* provide a useful starting point. Replenishment in fluid starved systems is essentially controlled by the amount, viscosity and surface tension of the free lubricant available *(10), (11)* and the time between overrollings. The film thickness is now determined by the quantity of lubricant available in the inlet *(10)* and the flow balance to and from the contact *(9), (11)-(13)*, rather than conventional EHL factors. Lubricant is lost from the track either due to fluid displacement in the inlet region or as squeeze flow from the contact *(10), (12)*. As the amount of lubricant is reduced then the first mechanism ceases to be important *(11)*. In addition as the EHL film thins fluid loss from the high pressure Hertzian zone becomes negligible *(10), (11)*. Thus, eventually, a flow balance is established between material lost from the track and replenished from the lubricant reservoir *(10)*.

Starved grease lubrication is analogous to the fluid film case, except there is the additional complication of a complex, time dependent rheology. In the literature, two different types of reflow are proposed for track replenishment:

1. Bulk redistribution: overrolling action of the ball pushes grease around contact *(4)* and 'finger' formation in the cavitation zone pulls grease towards the track *(8)*.
2. Local base oil reflow: surface/capillary driven flow around the contact of base oil from grease reservoir at side of track *(4)*.

It is possible that the bulk movement of the grease closer to the track helps to feed the local replenishment. The second mechanism includes spontaneous reflow *(10)* of lubricant from the reservoir at the side of the track and local capillary driven flow *(2), (4)*. Both are driven by surface tension gradients and retarded by viscous forces. Replenishment will also be affected by the amount of low viscosity material released from the grease. This is determined by the shear stability and oil release properties of the grease and the operating conditions (time, rolling speed, temperature). Previous work *(9)* has shown that the steady-state film thickness level of a grease correlates with oil bleeding values. The base oil viscosity and thickener content will thus determine both the oil release and reflow properties, and hence control the degree of starvation. At present it is not possible to predict starved film thickness from a knowledge of bulk grease properties and it this aspect which is considered in this study.

This paper reports a systematic study of the effect of grease properties on film formation in the starved regime. A series of simple lithium hydroxystearate greases have been tested in a rolling contact device which simulates bearing operation. Central film thickness is measured with running time, at constant rolling speed, so that an equilibrium level is established. The effect of base oil viscosity, thickener concentration, bulk temperature and rolling speed on starved grease lubrication have been examined.

EXPERIMENTAL PROGRAM

Grease lubrication behavior has been studied in a rolling contact EHL device supplied by PCS Instruments Ltd. Thin film optical interferometry is used to measure the average film thickness in the centre of the contact for a range of rolling speeds, temperatures and supply conditions.

Optical Interferometry Test Device

The thin film optical interferometry technique has been described in earlier papers *(2)*, therefore only a brief summary of the technique and experimental apparatus will be given here. The EHL contact is formed by a steel ball loaded against a glass disc which is driven by an electric motor. The underside of the glass disc is coated with a semi-reflecting chromium film and a silica spacer layer. A modified optical interferometry technique is used to measure film thickness in the centre of the contact. This technique can measure films down to 1 nm with a thickness resolution of 1nm. Typically three readings are taken at each speed level and the average value determined.

Two test procedures were used:

1. Fully flooded—film thickness measured with increasing and then decreasing speed. In these tests a small channeling device was used to push the overrolled grease back into the track.
2. Starved—film thickness change with time measured at constant speed. In these tests a single charge of grease was injected into the rolling track at the start of the test. Film thickness measurements are taken periodically over 60 minutes for the 0.1 m/s and 30 minutes for the 0.5 m/s tests, respectively.

The test conditions are summarized in Table 1. All tests were repeated to ensure representative results were obtained.

A matrix of six grease samples were tested with two base oil viscosities (30 and 200 cSt @ 40°C) and three thickener concentrations (5, 9 and 15%). All the samples came from the same supplier and were additive-free. The base oil was a paraffinic mineral oil and the thickener lithium hydroxystearate.

Base oil bleeding from the greases was also measured in a simple experiment to compare oil release at 80°C. Grease (10g) was placed in a filter paper cone supported by a funnel, this was then placed in a constant temperature oven at 80°C for 75 hours. The

TABLE 1—SUMMARY OF TEST CONDITIONS AND GREASE PROPERTIES			
TEST CONDITIONS		GREASE PROPERTIES	
Load	20N	Thickener	Lithium Hydroxystearate
Temperature °C	25 - 80	Thickener Content	5, 9, 15% w/w
Rolling Speed m/s	0.005-1.0	Base Oil Type	Paraffinic
Max Pressure GPa	0.5	Base Oil Viscosity	30,200 cSt @ 40°C

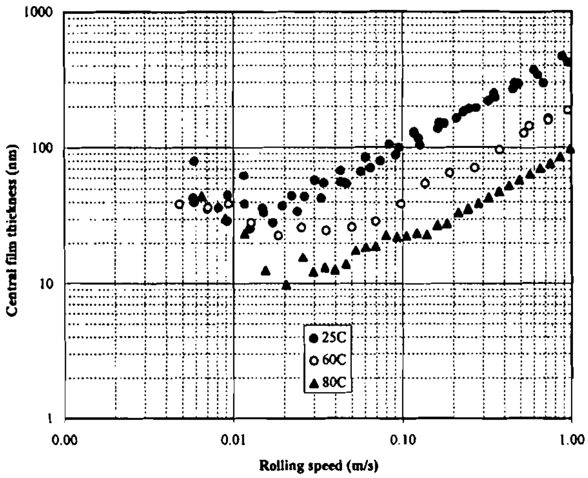

Fig. 1—Comparison of fully flooded film thickness results at different temperatures for 5% 30 cSt greases.

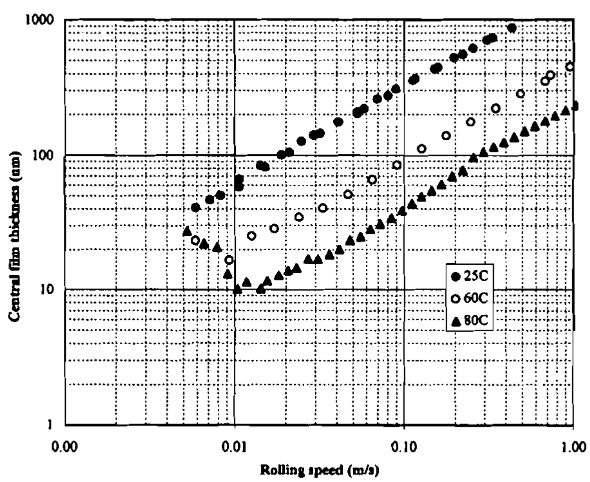

Fig. 2—Comparison of fully flooded film thickness results at different temperatures for 5% 200 cSt greases.

bled oil was collected and weighed.

FILM THICKNESS RESULTS

Fully Flooded results

Fully flooded curves for two of the greases at three temperatures are shown in Figs. 1 and 2. Generally, film thickness increases with rolling speed. However, at low rolling speeds greases often give unusually thick films, which decrease as the speed increases, contrary to usual fluid film behavior. This can be seen in the results for the 5 percent 30 cSt greases given in Fig. 1 and for the 5 percent 200 cSt grease at 80°C. This anomalous behavior is due to the passage of thickener lumps through the contact. In this region, measurements fluctuate wildly and the standard deviation (over the three readings taken for each measurement) is very high. This behavior often becomes more marked at higher temperatures and there are two possible explanations for this, first, that the shear stress in the inlet (due to base oil viscosity reduction) is lower at higher temperatures so the largest lumps are more likely to survive. Second, that the distortions are more obvious given the generally thinner EHL films.

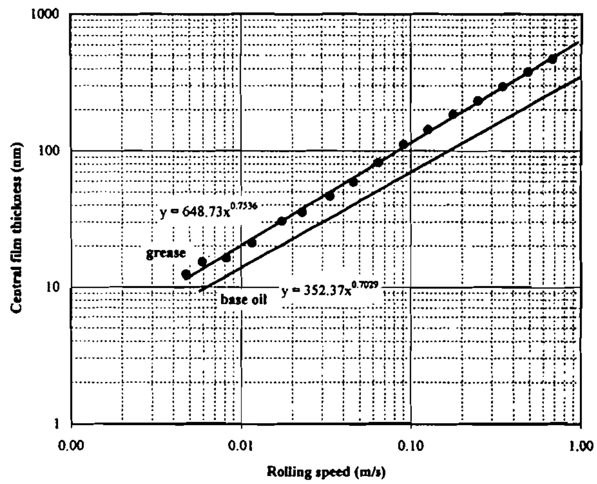

Fig. 3—Curve fitted film thickness results for 9% 200 cSt grease and base oil at 60°C.

With increasing speed film thickness often dropped, reached a minima and then climbed again. It was noticeable that at this speed transition point the nature of the film within the contact changed. Above the speed transition point, the image appears to be of a normal EHL film and the standard deviation is very low; often less than five nm. In this region film thickness increases with entrainment speed and the log-log film thickness-speed exponent is usually in the range 0.6 - 0.75. This behavior is very similar to that for the base oil and suggests that the rheology in the inlet region is now essentially Newtonian; this is demonstrated in Fig. 3 where the results for the 9 percent 200 cSt grease at 60°C are plotted together with those for the base oil. The results have been curve fitted (power law) and the resulting equations are displayed on the graph. The gradients are 0.75 for the grease and 0.7 for the base oil; these are in good agreement with the Dowson-Hamrock equation speed exponent of 0.68 *(5)*. Generally, the film thickness increases with increasing inlet viscosity (base oil viscosity and thickener content) and rolling speed, in agreement with classical EHL theory. The fully flooded results for all the greases are summarized in Table 2.

Starved Results

Representative film decay results are plotted as a function of disc revolution in Figs. 4 and 5 for 30 cSt and 200 cSt greases at 0.1 m/s and 25°C. In these tests the effect of increasing thickener content on starvation can be seen. The results for the 30 cSt greases show a clear progression in the final, equilibrium, film thickness decreasing with increasing thickener content as follows:

Table 2—Summary of Fully Flooded Test Results (in nm) at 0.1 and 0.5 m/s

Grease	25°C		60°C		80°C	
	0.1	0.5	0.1	0.5	0.1	0.5
5% 30 cSt	110	300	40	140	23	60
9% 30 cSt	120	340	40	130	20	55
15% 30 cSt	140	420	50	145	22	80
5% 200 cSt	350	950	100	300	40	158
9% 200 cSt	410	>1000	120	390	54	190
15% 200 cSt	450	>1000	150	490	60	210

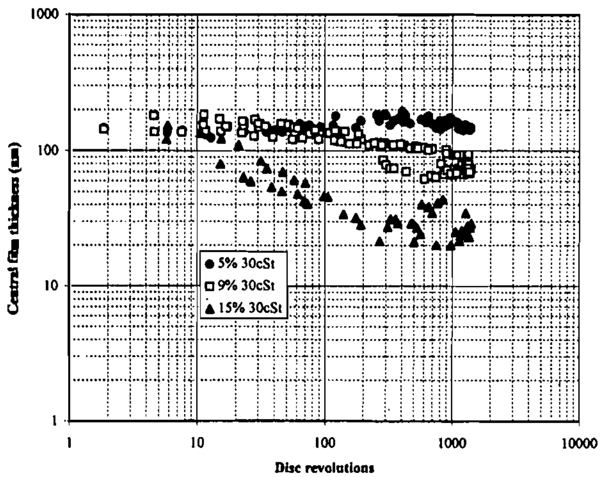

Fig. 4—Starved film decay for the 30 cSt greases at 0.1 m/s, 25°C.

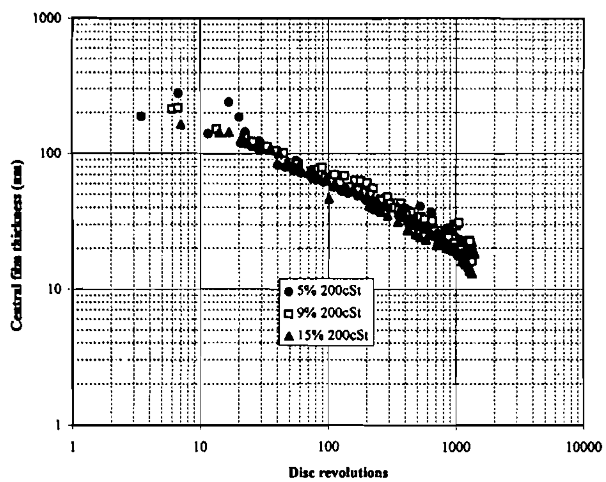

Fig. 5—Starved film decay for 200 cSt greases at 0.1 m/s 25°C.

5% > 9% > 15%
equilibrium starved film decreasing
→

The results for the five percent are very close to the fully flooded value indicating that the bulk viscosity of the grease is low enough for the overrolling action of the ball to maintain flow into the inlet. As the thickener content increases, however, the system

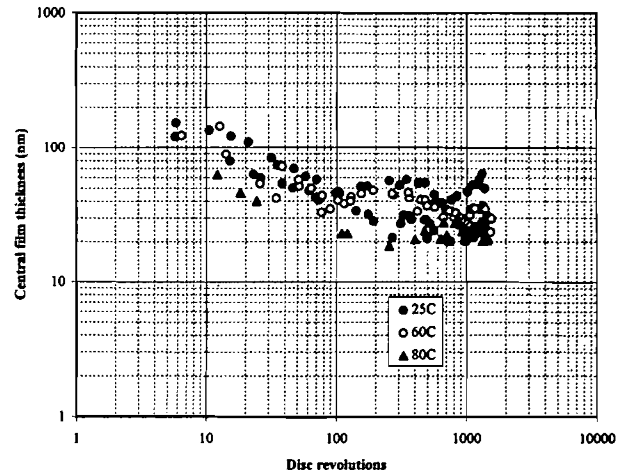

Fig. 6—Starved film thickness decay for 15% 30 cSt greases at 0.1 m/s.

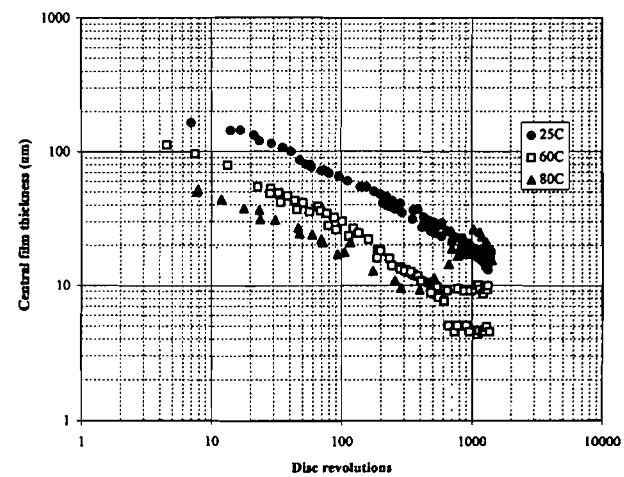

Fig. 7—Starved film thickness decay for 15% 200cSt greases at 0.1 m/s.

starved more readily. The results for the nine percent show a slight decrease before levelling-off at 80-90 nm whilst the 15 percent gives a much lower equilibrium film of 20-45 nm. This starved film thickness ranking would not have been predicted by the fully flooded results which increase with increasing thickener content as expected.

The starved results for the 200 cSt greases are very similar for all thickener concentrations. The film continues to decay throughout the test and there is little indication that an equilibrium condition has been established. Overall the equilibrium starved film thickness for the 200 cSt greases is less than for the 30 cSt greases, again this result would not have been predicted by normal EHL considerations.

In Figs. 6 and 7 the results of the 15 percent greases (30 and 200 cSt, respectively) are compared at different temperatures. The 30 cSt greases show similar behavior for all temperatures: a steady decay over the first 100 revolutions followed by a levelling off. The equilibrium film thickness is in the overall range 20-50 nm. The 200 cSt greases give a similar decay. However, in this case there is a clear distinction between the curves. Film decay for the 25°C case continues to the end of the test and for the 60°C lev-

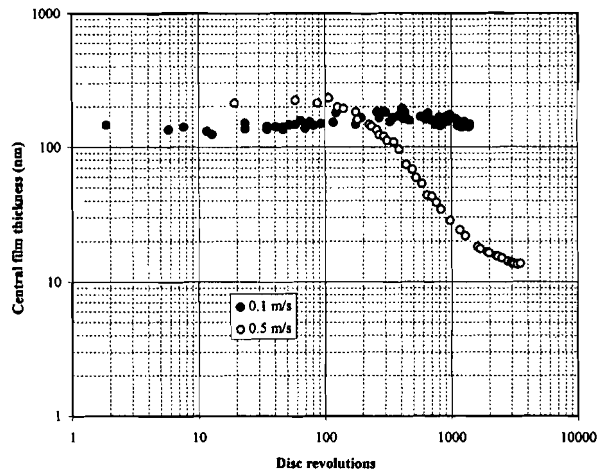

Fig. 8—Effect of rolling speed on starved film thickness 5% 30cSt greases at 25°C.

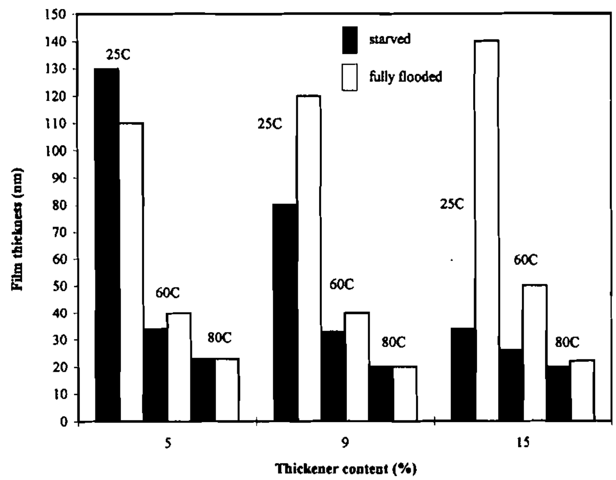

Fig. 9—Comparison of starved and fully flooded film thickness at 0.1 m/s for the 30 cSt greases.

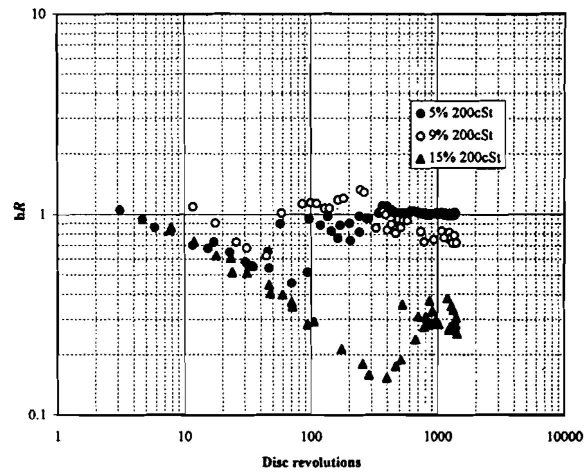

Fig. 10—Effect of thickener content on relative film thickness (h_R = starved film thickness/fully flooded film thickness) decay at 80°C, 200 cSt base oil, 0.1 m/s.

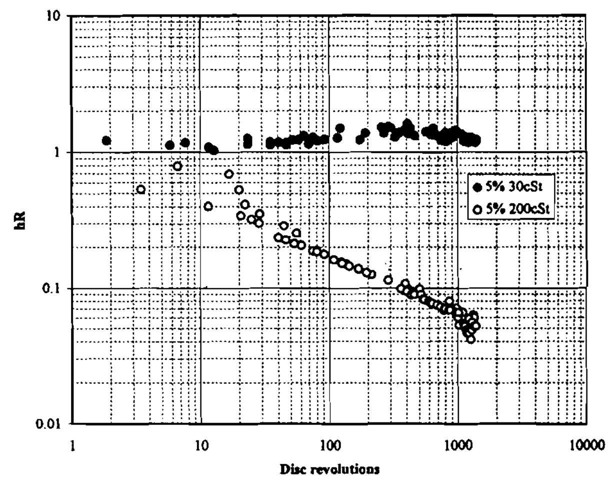

Fig. 11—Effect of base oil viscosity on relative film thickness decay (h_R = starved film thickness/fully flooded film thickness), 5% thickener, 0.1 m/s 25°C.

els off after 500 revolutions. The film thickness for the 80°C case levels off and then increases after 400 revolutions to give a final equilibrium value of 25-30 nm; which is higher than that obtained for the 25°C tests.

The effect of speed on starved film thickness is shown in Fig. 8 where results are plotted for the five percent 30 cSt grease at 25°C. This example was chosen to show the very different behavior that can occur at higher speeds. The five percent grease shows a transition from fully flooded at the lower speed to heavily starved at 0.5 m/s. Increasing the speed in this case decreases the equilibrium starved film thickness.

The preceding results demonstrate the complexity of starved grease lubrication and dependence of the equilibrium film thickness on the different grease and test parameters. Film thickness in the starved regime does not necessarily scale with the usual EHL parameters and thus any predictive rules will be very different. In some cases the influence of speed or base oil viscosity is the inverse of what would be expected from fully flooded theory and this is because it is the ability of the grease to replenish the contact that is the decisive factor in the starved regime. The dependence of starvation on grease properties is analyzed in more detail in the next section.

Analysis of Starved Results

It is interesting to compare the film thickness under fully flooded and starved conditions and this is shown in Fig. 9 where starved (equilibrium) and fully flooded film thickness (both at 0.1 m/s) are plotted against thickener content for the 30 cSt greases. At 25°C the starved film thickness is much lower than the fully flooded for the 9 and 15 percent greases. However, with increasing temperature the starved film thickness approaches that of the fully flooded value for all thickener contents. The apparent anomaly of the starved film being greater than the fully flooded value for the five percent grease at 25°C is due to continued flow of thickener lumps into the contact at the low rolling speed (see above). Generally, the results show that the degree of starvation increases as the rolling speed, oil viscosity and thickener content

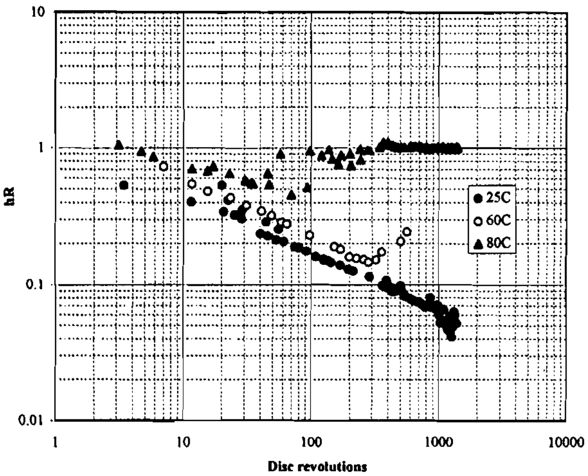

Fig. 12—Effect of temperature on relative film thickness decay (h_R = starved film thickness/fully flooded film thickness), 5% 200 cSt grease.

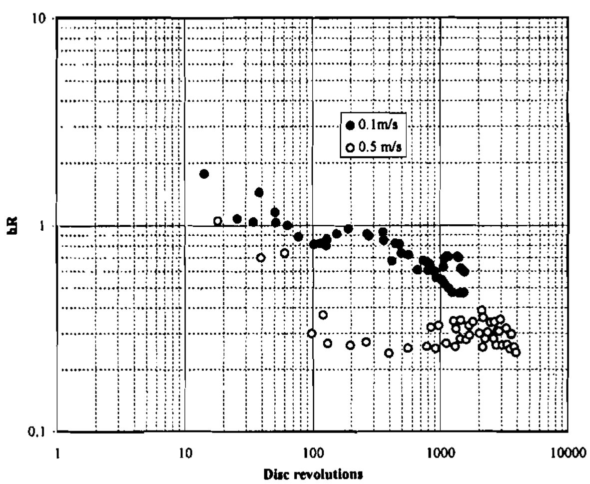

Fig. 13—Effect of rolling speed on relative film thickness decay (h_R = starved film thickness/fully flooded film thickness), 15% 30 cSt grease, 60°C.

increases.

This can be seen more clearly in the results plotted in Figs. 10-13, here film thickness is expressed as relative film thickness (h_R), which is a measure of the degree of starvation, where:

$$h_R = \text{starved film thickness/fully flooded film thickness}$$

The effect of increasing thickener content, base oil viscosity, temperature and rolling speed on the degree of starvation are shown in Figs. 10-13.

The film thickness in the starved regime depends critically on the amount of lubricant returning to the track, this will be determined by the amount of oil available for reflow, base oil viscosity and time between overrollings. The amount of oil available for reflow will be determined by the structural stability and oil release properties of the grease. In some cases the film decays to a low value before recovering towards the end of the tests suggesting

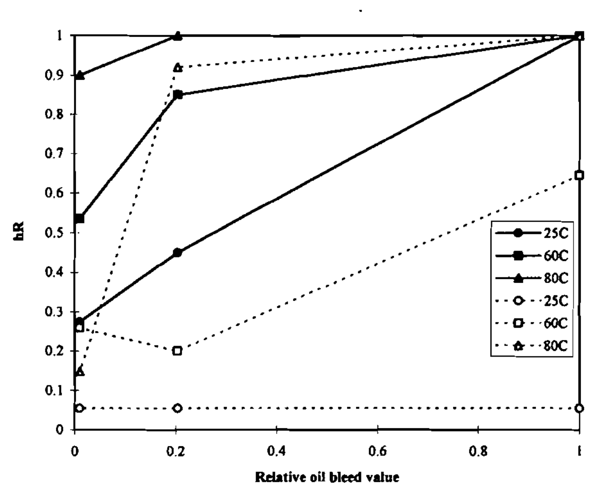

Fig. 14—Comparison of oil bleed and equilibrium film values for 30 cSt greases at 0.1 (closed symbols, solid line) and 0.5 (open symbols, dotted line) m/s.

that there is more oil available for replenishment due to the worked grease releasing base oil. The propensity of the grease to bleed oil might also be taken as an indication of oil release during use. The measured oil bleed results are summarized below where a relative ranking is given, the actual amounts of bled oil are shown in brackets below.

Oil bleed ranking:

5%30 > 5% 200 > 9%30 > 9%200 > 15%30, 15%200
(3.52g) (1.9g) (0.72g) (0.05g) (0), (0)

Overall oil release decreased with increasing thickener content and base oil viscosity; a trend which agrees with the starved film thickness results. This correlation is shown in Fig. 14 where the equilibrium h_R is plotted against oil bleed value for the 30 cSt grease for all test temperatures.

CONCLUSIONS

The current research program has concentrated on studying fundamental mechanisms of grease behavior in a simple rolling test device. Film thickness has been measured, under both fully flooded and starved conditions, for a series of simple, lithium hydroxystearate thickened greases. The effect of bulk grease properties and test conditions on starved film thickness has been demonstrated. The conclusions from this work are summarized below:

1. Fully flooded film results: Film thickness scales with the usual EHL parameters—increases with thickener content and base oil viscosity; at low rolling speeds thickener lumps pass through the contact distorting the EHL film. This effect could be related to the noise characteristics of the grease and at higher rolling speed film thickness increases ~ $(U)^{0.7}$ suggesting Newtonian behavior in the inlet region.
2. Starved film results: The starved film results have shown that the factors that determine film thickness in the fully flooded

regime can have the opposing effects in the starved regime; the degree of starvation increases with increasing base oil viscosity, thickener content (decreasing oil bleed) and rolling speed. It decreases with increasing temperature and in many cases film thickness drops to a fraction of the fully flooded value - particularly at rolling speeds approaching those of real bearing applications. Taking the fully flooded oil film thickness as representative of grease lubricant films in a bearing is therefore a gross overestimation.

The problem of formation and maintenance of grease lubricating films is essentially one of flow balance and the prevailing loss and supply mechanisms can change with test condition and grease properties. The factors that determine film thickness will differ in the two regimes. If full flow can be assumed then film thickness will scale with the usual EHL parameters of inlet rheology and entrainment speed. In the starved regime it is the availability of oil for reflow and the ability to replenish the contact which will determine film thickness.

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REFERENCES

(1) Wilson, A. R., "The Relative Thickness of Grease and Oil Films in Rolling Bearings," in *Proc. I.MechE*, **193**, pp 185-192, (1979).

(2) Cann, P. M., "Starvation and Reflow in a Grease Lubricated Elastohydrodynamic Contact," *Trib. Trans.*, **39, pp** 698-704, (1996).

(3) Booser, E. R. and Wilcock, D. F., "Minimum Oil Requirements of Ball Bearings," *Lubr. Eng.*, **9**, pp 140-143, pp 156-158, (1953).

(4) Åström, H., Ostenson, J. O. and Höglund, E., "Lubricating Grease Replenishment in an Elastohydrodynamic Point Contact," *ASME Jour. of Trib.*, **115**, pp 501-506, (1993).

(5) Hamrock, B. J., "Fundamentals of Fluid Film Lubrication," McGraw-Hill Int'l. Editions, (1994).

(6) Kauzlarich, J. J. and Greenwood, J. A., "Elastohydrodynamic Lubrication with Herschel-Bulkley Model Greases," *ASLE Trans.*, **15**, pp 269-277, (1972).

(7) Zhu, W. S. and Neng, Y. T., "A Theoretical and Experimental Study of EHL Lubricated With Grease," *Jour. Trib. ASME Trans.*, **110**, pp 38-43, (1988).

(8) Larsson, P.-O., "Lubricant Replenishment in the Vicinity of an EHD Contact," Ph.D. Thesis, Lulea University of Technology, (1996).

(9) Cann, P. M. E., Lubrecht, A. A. and Chevalier, F., "Track Depletion and Replenishment in a Grease Lubricated Point Contact: A Quantitative Analysis," in *Proc. 23rd Leeds-Lyon Symp., Elastohydrodynamics '96*, pp 405-413, (1997).

(10) Chevalier, F., Lubrecht, A. A., Cann, P. M.E., Dalmaz. G. and Colin, F., "Film Thickness in Starved EHL Point Contacts," *ASME JoT*, **120**, pp 126-133, (1998).

(11) Chiu, Y. P., "An Analysis and Prediction of Lubricant film Starvation in Following Contact Systems," *ASLE Trans.*, **17**, pp 22-35, (1974).

(12) Guangteng, G., Cann, P. M. and Spikes, H. A., "A Study of Parched Lubrication," *Wear*, **153**, pp 91-105, (1992).

(13) Kingsbury, E., "Parched Elastohydrodynamic Lubrication," *Trans. ASME Jour. Trib.*, **107**, pp 229-233, (1985).